



**Year 11 Mathematics**  
**Preliminary ASSESSMENT TASK 3**  
**TERM 2 Week 6 2003**

Set by: SM

Tuesday 3rd June, 2003

- Attempt **ALL** questions.
- Marks may be deducted for careless, insufficient, or illegible work.
- Calculators may be used.
- Begin a new sheet of paper when you are instructed to do so on the paper.
- Not all questions are of equal value.
- Total possible mark is 60.

**TIME ALLOWED:** 80 minutes

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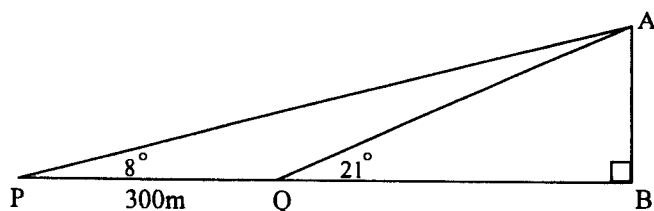
**Question 1 :**

- (a) Evaluate  $\sin 63^\circ 47'$  to 3 decimal places. [1]
- (b) Given that  $\tan \theta = \frac{1}{4}$  and that  $\theta$  is acute, find the exact value of  $\cos \theta$  and  $\operatorname{cosec} \theta$ . [2]
- (c) Find the value of  $\theta$  if  $\sin(\theta + 25^\circ) = \cos 35^\circ$  [2]
- (d) Simplify the expression,  $\frac{\operatorname{cosec} 10^\circ}{\sec 80^\circ}$  [1]
- (e) Show that the expression  $\frac{\sin 45^\circ}{\cos 30^\circ \cdot \tan 60^\circ} = \frac{\sqrt{2}}{3}$  [3]
- (f) Find the equivalent acute angle expression for  $\cos 200^\circ$ . [2]
- (g) If  $\cos \theta = -\frac{2}{5}$  and  $\sin \theta < 0$ , find the exact value of  $\tan \theta$ . [2]
- (h) Solve the following trigonometric equations given that  $0^\circ \leq \theta \leq 360^\circ$ . Where necessary, give your answer to the nearest degree.
- |                                  |                                           |
|----------------------------------|-------------------------------------------|
| (i) $\tan \theta = -\frac{2}{5}$ | (iii) $\cos^2 \theta = \frac{3}{4}$ [2,2] |
| (ii) $2\sin 2\theta = 1$         | (iv) $\sin \theta = 0$ [2,1]              |

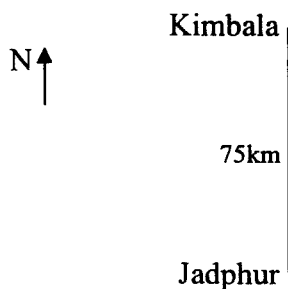
**Question 2 : (BEGIN A NEW PAGE)**

- (a) Simplify the expression,  $\cot \theta \sin \theta$ . [1]
- (b) Simplify the expression,  $\sec \theta \operatorname{cosec} \theta \tan \theta$ . [2]
- (c) Prove that  $\tan^2 \theta \cos \theta + \cos \theta = \sec \theta$  [3]

- (d) From position P, John finds the angle of elevation of the top of a tower, AB, to be  $8^\circ$ . After walking 300m directly towards the tower to point Q, he finds the angle of elevation to be  $21^\circ$ .



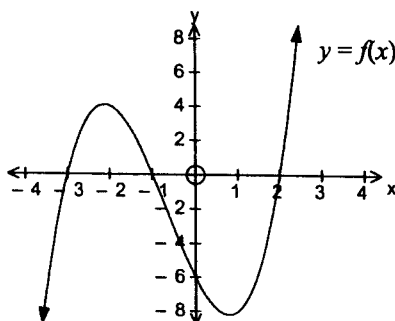
- (i) Explain why  $\angle PAQ = 13^\circ$ . [1]
- (ii) Calculate the length of AQ (to 1 decimal place) [3]
- (iii) Find the height of the tower to the nearest metre. [2]
- (e) Jadphur is a town 75km directly south of the town Kimbala, as shown on the diagram. The town Lati is 50km from Kimbala on a bearing of  $125^\circ$ .
- (i) Copy the diagram below and mark in the position of Lati. There should be at least one angle on your diagram [2]



- (ii) Find the distance of Lati to Jadphur to the nearest kilometre. [3]
- (f) (i) On the same set of axes, sketch the graphs of  $y = \sin x$  and  $y = \tan x$  for  $0^\circ \leq x \leq 360^\circ$ . [2]
- (ii) By using your graph, or otherwise, predict the number of solutions to the equation  $\sin x = \tan x$  for  $0^\circ \leq \theta \leq 360^\circ$ . [2]

**Question 3 : (BEGIN A NEW PAGE)**

- (a) If  $y = 5$ , find  $\frac{dy}{dx}$ . [1]
- (b) Find the derivatives of the following functions
- (i)  $y = 4x^4 - 3x^2 + x - 1$  [1]
- (ii)  $y = \frac{1}{x} - \frac{3}{x^2}$  [2]
- (iii)  $y = \sqrt[3]{x^2}$  [2]
- (c) Find the gradient of the tangent to the graph of  $y = x^3 - 2x^2 - 1$  at the point where  $x = 2$ . [2]
- (d) Find the equation of the normal to the graph of  $y = \frac{x^2}{2} - 1$  at the point where  $x = -5$  [4]
- (e) The graph of the function  $y = f(x)$  is shown below.



- (i) Copy the diagram onto your answer sheet and on the same axes, sketch  $y = f'(x)$ . [2]
- (ii) For what values(s) of  $x$  is  $f'(x) < 0$  ? [2]
- (f) Consider the function  $f(x) = \frac{1}{3}x^3 - 2x^2 + 1$ . For what value(s) of  $x$  is the tangent to the curve horizontal ? [3]

**END OF TASK**